

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

4. Let $G = (V, E)$ be a weighted graph with $V = \{0, 1, 2, 3, 4, 5\}$ and the edge between i and j with weight w represented by the triple (i, j, w) in $E = \{(0, 5, 5), (1, 3, 8), (1, 2, 6), (2, 5, 5), (0, 2, 1), (3, 5, 4), (4, 5, 7), (3, 4, 5), (1, 4, 5)\}$.

- (a) Construct the minimum spanning tree with the reverse-delete algorithm. Show all steps.
(b) For which type of graph would you not use the reverse-delete algorithm and use Prim or Kruskal? Justify your answer by referring to the asymptotic cost of the algorithms.

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Answer:

4. (a) i. delete $(1, 3, 8)$
ii. delete $(4, 5, 7)$
iii. delete $(1, 2, 6)$
iv. delete $(0, 5, 5)$

We are left with $\{(0, 2, 1), (2, 5, 5), (3, 5, 4), (3, 4, 5), (1, 4, 5)\}$, 5 edges connecting 6 vertices, and thus a spanning tree.

- (b) If the graph is fully connected, $m = n(n - 1)/2$, then the algorithm will have to delete $m - (n - 1)$ edges to arrive at a tree and thus run through many steps.

Prim's and Kruskal's algorithm run in $n - 1$ steps to connect all n vertices. However, there asymptotic cost is also $O(m \log(n))$, where m is $O(n^2)$ for a fully connected graph.